Acute Lymphoblastic Leukaemia in Children and Adolescents

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73.1 Introduction

Acute lymphoblastic leukaemia (ALL) is the most common paediatric cancer; approximately 60% of ALL cases occur in children and adolescents younger than 20 years (Inaba and Pui 2021). Allogeneic haematopoietic cell transplantation (HCT) became the most commonly applied cellular immunotherapy and the standard of care for children with ALL who are either at high risk for relapse or had previously experienced a disease recurrence. HCT represents a successful therapeutic option and a relevant proportion of patients achieved long-term survival (Balduzzi et al. 2019). The most frequent cause of treatment failure is relapse after allogeneic HCT. The risk of post-transplant relapse is influenced by several factors, including remission status at transplantation, conditioning regimen and donor type (Peters, et al. 2021). Strategies aimed at reducing the risk of treatment failure have included a reduction of minimal residual disease (MRD) pre-transplant (Bader et al. 2019), the substitution of toxic chemotherapy before transplantation with bispecific antibodies (Locatelli et al. 2021), replacement of HCT with chimeric antigen receptor (CAR) T-cell therapy (Maude et al. 2018a, b), improved transplant strategies for specific groups, including infants (Pieters et al. 2019), adolescents and young adults (AYA) (Diesch-Furlanetto et al. 2021) and innovative prophylaxis and treatments for acute and chronic graft-vs.-host disease (GvHD). Furthermore, therapeutic drug monitoring with dose adjustment of some drugs, including busulfan (Diesch-Furlanetto et al. 2021) and novel radiation techniques might enable a more individualised approach (Hoeben et al. 2021).

To offer the patients the best available treatment options, a close collaboration between international study groups and transplant consortia is necessary. As an example, this cooperation was realized within the treatment consortia for childhood leukaemia (e.g. IBFM-SG, IntReALL, NOPHO, UKALL, AIEOP, FRALLE and others) and the paediatric transplant community (e.g. EBMT-PD WG, IBFM-SC SCT,

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GETMON and GITMO). ALL trial groups assess outcome according to their chemotherapy protocols and stratify patients into standard, intermediate and high relapse risk groups. In contrast to adults, only high-risk patients are eligible for allo-HCT in first complete remission (CR).

73.2 Prognostic Factors and Indications for HCT

HCT indications have to be defined prospectively and must be re-evaluated and regularly revised at intervals dependent on modifications and improvements in non-transplant approaches for both front-line and relapse protocols. Some risk factors such as recurrent molecular lesions predicting a poor outcome conveying a dismal prognosis in childhood ALL can be identified at diagnosis (Moorman 2016; O'Connor et al. 2018). In addition, the response to induction therapy measured by MRD levels has a strong prognostic value and now defines many indications for HCT (Bader et al. 2009; Conter et al. 2010; Schrappe et al. 2011; Eckert et al. 2013; Berry et al. 2017).

73.2.1 Indications: CR1

Only patients with high-risk cytogenetic features or unsatisfactory response to chemotherapy are eligible for HCT in first remission. In contrast to earlier recommendations, for these patients an MSD and an MUD and for the highest relapse category also mismatched donors (MMD) are an option (Truong et al. 2021). In the ongoing AIEOP-BFM 2017-trial, the very-high-risk subgroup with an indication for HCT in first CR is defined by

1. The presence of TCF3-HLF gene fusion,
2. KMT2A-AFF1 gene fusion,
3. Hypodiploidy defined as <44 chromosmes,
4. IKZF1^{plus} deletions and medium-risk/high-risk MRD levels at the end of consolidation,
5. PCR-MRD HR, and.
6. T-ALL with PPR and/or FCM-MRD d15 HR and/or IF.

Patients with MRD negativity at EOI are excluded from a HCT indication. MMD-HCTs are nowadays reserved for a PCR-MRD TP2 $\geq 5 \times 10^{-3}$, all TCF3-HLF fused leukaemias and those with induction failure (Table 73.1).

Table 73.1 Indications for allogeneic HCT in CR1 according to AIEOP-BFM ALL 2009-trial

		PCR-MRD results[a]				
				MRD-HR		
		MRD-SR	MRD-MR[b]	MRD TP2 $\geq 10^{-3}$ to $<10^{-2}$	MRD TP2 $\geq 10^{-2}$	No MRD result
Criteria hierarchical	No CR d33	No[c]	MMD	MMD	MMD	MMD
	t(4;11)[d]	No	MD	MD	MMD	MD
	Hypodiploidy <44 chromosomes[e]	No	MD	MD	MMD	MD
	PPR + T-ALL	No	No	MD	MMD	MD
	None of the above features[f]	No	No	MD	MMD	No

PPR Prednisone Poor Response on day 8, *NRd33* No Remission on day 33 *MRD* Minimal Residual Disease, *no* Allo HCT not indicated, *MD* Permitted donor: HLA-matched sibling or non-sibling donor, *MMD* Permitted donor: HLA-matched or HLA-mismatched donor

[a] FCM-MRD results have no impact on the allo-HCT indication

[b] Including MRD-MR SER (MRD TP1 $\geq 10^{-3}$ and TP2 $10^{-4/-5}$)

[c] Non-remission in patients with this rare constellation should be due to extramedullary disease. Allo-HCT indication in these cases should be discussed with the national study coordinator

[d] Independent of prednisone response

[e] The finding of exactly 44 chromosomes qualifies for HR treatment but has no impact on allo-HCT indication

[f] Including patients with 44 chromosomes

Table 73.2 Indication for HCT according to IntReALL SR 2010 and HR protocol criteria

Relapse risk group	Phenotype	Time of relapse	Site of relapse	MRD-status	Donor type
Very high	T-ALL	Any time	I-BM, C-BM, I-EM		MSD, MD, MMD
	Non-T-ALL				
		Very early	I-BM, C-BM, I-EM		
		Early	I-BM, C-BM	PR, ND	
High	Non-T-ALL				MSD, MD
		Late	I-BM, C-BM	PR, ND	
		Early	C-BM	GR	
			I-EM		
		Late	I-BM	PR, ND	
			C-BM	ND	

I-BM isolated bone marrow, *C-BM* combined bone marrow and extramedullary site, *I-EM* isolated extramedullary, *MRD: GR* good response as defined by the specific chemotherapy-protocol, *PR* poor response, *ND* not detectable

73.2.2 Indications: CR2 and Later

All patients with relapse of T-ALL and patients with B-ALL who relapse during frontline therapy or within 6 months from treatment discontinuation (very early and early relapse) have a dismal prognosis when treated with conventional chemotherapy. Allo-HCT from any donor type is the contemporary standard post-remission consolidation therapy (Table 73.2a).

In multiple relapsed patients, if they achieve a third or higher remission, allo-HCT should be considered in all patients, provided that their condition allows such a procedure. Patient not in morphological remission should not be transplanted except in extraordinary experimental situations (e.g. well-designed clinical trial with clear scientific questions).

73.3 Donor Selection and Stem Cell Source

OS and incidence of NRM in patients transplanted from a MSD as well as from a > 9/10 MUD have steadily improved over time, and today the results for these outcomes do not differ according to the type of donor. However, it has been shown that in children HCT from an HLA-identical sibling results in a faster myeloid engraftment, prompter immunoreconstitution and less severe infections and should remain the preferred option (Peters et al. 2015a, b).

Since less than 25% of patients have a MSD, HCT from an alternative donors is most frequently applied. Several groups have demonstrated that an HCT from a MUD, identified by HLA high-resolution typing, has a similar outcome as a MSD-HCT (Zhang et al. 2012; Fagioli et al. 2013; Burke et al. 2015) Notably, Dalle et al. recently reported the 3-year consolidated results of 612 patients aged 4–21 years, treated at >100 pediatric centers in 26 countries following HCT from either MSD (n = 186, 30%) or MUD (defined as 9 or 10/10 4-digit molecular HLA compatibility, n = 426, 70%) following 12Gy TBI-VP16 for ALL in CR1. GvHD-relapse-secondary malignancy-free survival (GRFS) was significantly better in MUD vs. MSD (62 ± 3 vs. ±51 ± 5%, P = 0.04) (Dalle et al. 2022).

Several methods were developed to overcome the HLA barriers in those patients who lack an HLA compatible donor. Currently, it is not clear which of the following, HLA-mismatched CB, TCD (alpha-beta depleted, CD34+ selected or CD3/CD19 depleted) haploidentical grafts or PT-CY approaches, will result in the best outcome (Lang and Handgretinger 2008; Smith et al. 2009; Ruggeri et al. 2014; Locatelli et al. 2017; Rocha, et al. 2021) (Tables 73.3 and 73.4). Very promising results have been recently confirmed in children with ALL receinving a graft from an α/β T cell depleted HLA-haploidentical relative, with a very low risk of NRM and of both acute and chronic GvHD (Merli et al. 2022).

Table 73.4 Donor hierarchy—further selection criteria

Variable/ order	Priority
CMV-status	
Patient CMV IgG positive	
1	Donor CMV IgG positive
2	Donor CMV IgG negative
Patient CMV IgG negative	
1	Donor CMV IgG negative
2	Donor CMV IgG positive
Gender	
(Locatelli et al. 2022)	
1	Male or female (preferentially not Allo-immunized by prior pregnancy) donor
Male patient	
1	Male donor
2	Female (preferentially not Allo-immunized by prior pregnancy) donor
Age	
1	Younger donor if body weight enables sufficient SC harvest
2	Older donor
Stem cell source	
HCT from MSD or MD	
1	Bone marrow
2	PPBSC (CAVE: Adjust GvHD-prophylaxis for matched siblings)
2	Cord blood with sufficient cell number (>3 × 10^7 NC/kg)
HCT from MMD: Possible options	
	BM, 8/10 matches, unmanipulated
	PBSC, haploidentical, CD3/CD19 depleted, α/β depleted
	CB, sufficient stem cell dose
	PBSC, haploidentical, CD34+ selection
	PT-CY

Table 73.3 Matching criteria according to HLA typing/ matching and stem cell source for children and AYAs with ALL

MSD	HLA-genotypically matched sibling, or 10/10 allelic match (if parental haplotypes unknown)	BM, PBSC
MSD	6/6 or 8/8, 5/6 or 7/8[a]	CB
MD	9/10 or 10/10 allelic matched related or unrelated	BM, PBSC
MD	5–6/6 unrelated or 6–7-8/8 unrelated	CB
MMD	Less than 9/10 matched	BM, PBSC
MMD	Less than 5/6 or 6/8 UCB	CB

MSD matched sibling donor, *MD* matched donor, *MMD* mismatched donor

[a] 4 digits high-resolution typing recommended also for CB matching definition

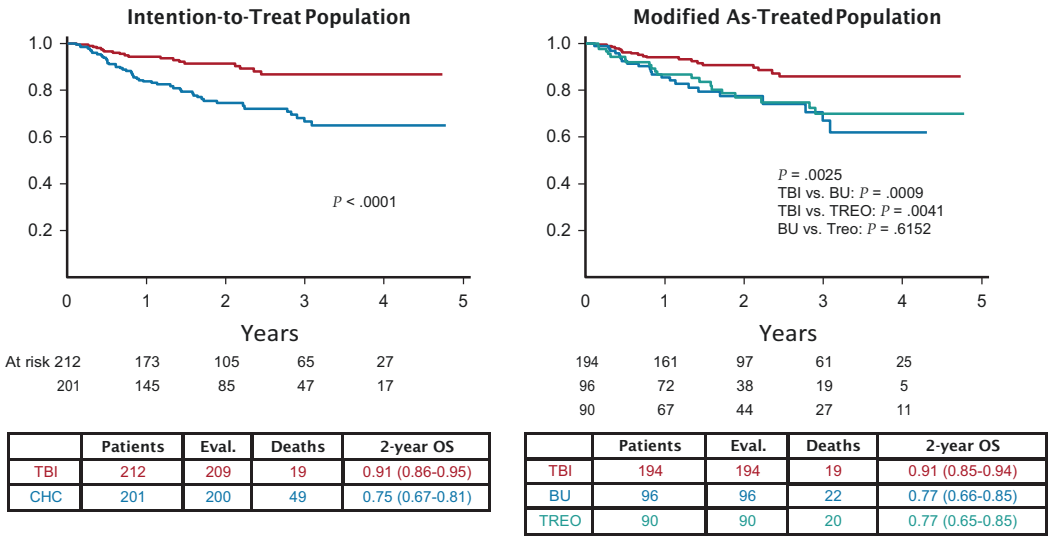
Other factors to be considered in the choice of the donor are detailed in Tables 73.3 and 73.4. In particular, the donor/recipient CMV serology plays a relevant role in the post-transplant outcome as well as the donor age.

73.4 Conditioning Regimen

A myeloablative conditioning regimen is the treatment of choice for children and AYA with ALL. The FORUM trial demonstrated the significant superiority of the TBI/etoposide scheme in a randomized, prospective, multicenter, phase III trial. At a median age of 4.5 years, patients over 4 years of age with high-risk ALL had a 20% better OS and EFS and a lower NRM compared to patients who received a chemotherapy-based conditioning regimen.(Locatelli et al. 2022). In addition, the risk of leukaemia recurrence was much lower in patients given TBI as part of the conditioning regimen in comparison to those prepared with a chemo-therapy-based myeloablation. The same study showed that a myeloablative regimen containing treosulfan, thiotepa and fludarabine was associated with an outcome comparable to that of patients treated with busulfan, thiotepa and fludarabine (Peters, et al. 2021) (Fig. 73.1).

Whether a myeloablative conditioning with treosulfan will have a more benign adverse event profile and a higher chance for fertility preservation remains to be proven (Wachowiak et al. 2011; Boztug et al. 2015; Lee et al. 2015; Faraci et al. 2019).

Whenever possible, the interval between the end of the last chemotherapy and the start of the conditioning regimen should be 3–6 weeks to reduce the risk of NRM. If infection or toxicity requires a delay in conditioning, patients receive risk-adjusted chemotherapy or immunotherapy to bridge the time to transplant.The most successful approach to allow a reduction of MRD before transplantation is the use of blinatumomab. The continuous infusion of the bispecific antibody enables not only a deeper remission but also to spare chemotherapy-associated organ toxicity which results in better pre-transplant performance status (Locatelli et al. 2000a, b).



	Patients	Eval.	Deaths	2-year OS
TBI	212	209	19	0.91 (0.86-0.95)
CHC	201	200	49	0.75 (0.67-0.81)

	Patients	Eval.	Deaths	2-year OS
TBI	194	194	19	0.91 (0.85-0.94)
BU	96	96	22	0.77 (0.66-0.85)
TREO	90	90	20	0.77 (0.65-0.85)

Primary end point: Overall survival. BU, busulfan; CHC, chemo-conditioning; CIR, cumulative incidence of relapse; EFS, event-free survival; OS, overall survival; TBI, total body irradiation; TREO, treosulfan; TRM, treatment-related mortality.

Fig. 73.1 Outcome of the randomized, prospective, multicentre phase III FORUM-study: (© 2020 by American Society of Clinical Oncology)

73.5 GVHD Prophylaxis

Children transplanted with bone marrow from MSD might benefit from an augmented GVL effect if only cyclosporine A is administered as GVHD prophylaxis (Locatelli et al. 2000a, b; Peters et al. 2010). However, careful monitoring and rapid intervention are crucial to prevent the development of severe GVHD. After HCT from non-sibling donors, a combination of a calcineurin inhibitor with short-course MTX and ATG is given in most patients (Peters et al. 2015a, b; Veys et al. 2012a, b; Balduzzi et al. 2019).

73.6 Children below 4 Years of Age

Both relapse and NRM contribute to treatment failure in infants and young children with high-risk ALL undergoing HCT. The optimal chemotherapeutic approach able to improve event-free survival (EFS) and to reduce the risk of NRM is not yet defined (Peters et al. 2022).

73.6.1 Mixed Chimerism (MC) and MRD

Mixed chimerism (MC) and MRD strongly predict risk for relapse in children (Bader et al. 2015).

Preemptive immunotherapy, e.g. withdrawal of IS or DLI guided by chimerism and MRD monitoring, may prevent impending relapse. However, the dynamic of leukaemic reappearance hampers the final success of these methods. Therefore, new post-transplant intervention strategies with less risk for severe complications like bi-specific antibodies or CAR-T-cell interventions may expedite the control of impending relapse (Handgretinger et al. 2011; Maude et al. 2018a, b).

73.6.2 Children with Ph + ALL

Children with Ph + ALL should receive TKIs post-transplant for relapse prevention. Whether the prophylactic approach (all Ph + patients will

receive TKIs) or a preemptive therapy (only patients with a persistence/reappearance of BCR/ABL fusion transcript) is more effective has yet to be demonstrated (Schultz et al. 2010; Bernt and Hunger 2014). Both strategies are currently under investigation.

73.6.2.1 The Amended EsPhALL Recommendation

Administration of imatinib prophylaxis post HCT when more than 50,000 platelets are reached is recommended with a duration of 365 days after HCT.

73.6.2.2 TKI According to MRD Result

Administration of imatinib post HCT for all MRD-positive patients until two negative results are achieved. FACS- and PCR-MRD analyses are accepted.

Key Points

- Only children and adolescents with very high or high relapse risk should be candidates for allo-HCT in CR1 and CR2. The assessment of relapse risk is largely influenced by the presence of recurrent molecular and cytogenetic abnormalities, response to chemotherapy, assessed through MRD evaluation and in relapsed patients – by time and site of relapse.
- MRD levels pre- and post-HCT are powerful predictors for outcome after HCT.
- Patients who are not in morphological remission before conditioning should not undergo allogeneic HCT except in extraordinary situations.
- Fractionated TBI/VP16-MAC is recommended for all children above the age of 4 with ALL. If this is not possible, a myeloablative chemo-conditioning is an option.

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